Theory Q4
Magnetic separation (8 points),
Solution and Marking Scheme

A.1
1.1pt

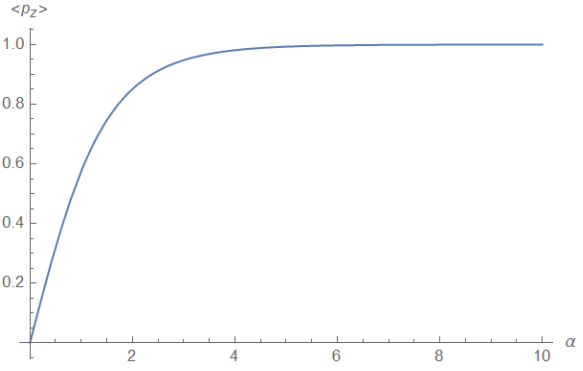
The energy of a dipole is $U_B = -\vec{p}\cdot\vec{B} = -p_z B$. Normalization: $A\sum_{m=-j}^{j} e^{\alpha m} = Ae^{-\alpha j}\frac{e^{(2j+1)\alpha}-1}{e^{\alpha}-1} = A\frac{\sinh(j+\frac{1}{2})\alpha}{\sinh\frac{\alpha}{2}} = 1$. Then $\langle p_z\rangle = Ag_j\mu_B\sum me^{\alpha m}$ with $\sum me^{\alpha m} = \frac{\mathrm{d}}{\mathrm{d}\alpha}\sum e^{\alpha m}$, which gives

$$\langle p_z\rangle = g_j\mu_B\left[\left(j+\tfrac{1}{2}\right)\coth\left(\left(j+\tfrac{1}{2}\right)\alpha\right) - \tfrac{1}{2}\coth\tfrac{\alpha}{2}\right].$$

$\langle p_z\rangle = g_j\mu_B\sum me^{\alpha m}/\sum e^{\alpha m}$	**0.2**
$\sum_{-j}^{j} e^{\alpha m} = \sinh\frac{\alpha(2j+1)}{2}/\sinh\frac{\alpha}{2}$	**0.3**
$\sum me^{\alpha m} = \frac{\partial}{\partial\alpha}\sum e^{\alpha m}$	**0.2**
The derivative evaluated correctly	**0.3**
Final formula	**0.1**

A.2
1.1pt

With $\coth x \simeq 1/x + x/3$: $\langle p_z\rangle \simeq \frac{j(j+1)}{3}\alpha g_j\mu_B$ for $\alpha \to 0$. With $\coth x \simeq 1 + e^{-2x}$: $\langle p_z\rangle \simeq g_j\mu_B\left(j - \frac{1}{2}e^{-\alpha}\right) \to jg_j\mu_B$ for $\alpha \to \infty$ ($j \neq 0$). The graph rises linearly from 0 and saturates at $jg_j\mu_B$:



$\langle p_z\rangle = \frac{g_j\mu_B j(j+1)}{3}\alpha$ for $\alpha \to 0$	**0.3**
$\langle p_z\rangle = g_j\mu_B j$ for $\alpha \to \infty$	**0.2**
Graph: monotonic increase	**0.2**
Graph: proportional at small α	**0.2**
Graph: saturation at large α	**0.2**

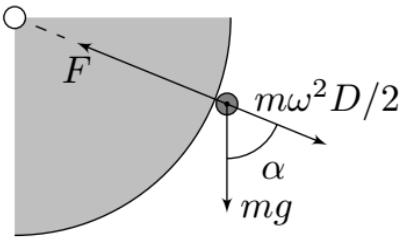
A.3 **0.9pt**	By definition $M = n\langle p_z\rangle$, so $$\chi \simeq \frac{j(j+1)}{3}\,\frac{n\mu_0 g_j^2 \mu_B^2}{kT}.$$	
	$\chi = \mu_0 n\langle p_z\rangle / B$	**0.1**
	Final formula $\chi = \frac{\mu_0 n g_j^2 \mu_B^2 j(j+1)}{3kT}$	**0.8**

B.1 **0.8pt**	$\ln(B/1\,\mathrm{T}) = \ln(B_0/1\,\mathrm{T}) - r/a$ is linear in r. A least-squares fit gives slope $k = -1/a = -0.445\ \mathrm{mm}^{-1}$ and intercept $b = \ln(B_0/1\,\mathrm{T}) = -0.012$, so $$B_0 = 1.0\ \mathrm{T}, \qquad a = 2.25\ \mathrm{mm}.$$	
	Linearization $\ln(B/1\,\mathrm{T}) = \ln(B_0/1\,\mathrm{T}) - r/a$	**0.2**
	Slope $k = -1/a \approx -0.445\ \mathrm{mm}^{-1}$	**0.2**
	Intercept $b = \ln(B_0/1\,\mathrm{T}) \approx -0.012$	**0.2**
	$a \approx 2.25$ mm	**0.1**
	$B_0 \approx 1$ T	**0.1**

B.2 **0.8pt**	The moment of the particle is $P = VM = V\chi_1 B/\mu_0$, so $\mathrm{d}E = -V\frac{\chi_1}{\mu_0} B\,\mathrm{d}B$ and $$E = -\frac{V\chi_1 B^2}{2\mu_0}.$$	
	$P = MV = V\chi B/\mu_0$	**0.2**
	Final formula $E = -\frac{\chi V B^2}{2\mu_0}$ (with a wrong sign or without the factor 2: 0.3 instead of 0.6)	**0.6**

B.3
1.1pt

The force on a particle is $F = -\frac{\mathrm{d}E}{\mathrm{d}r} = -\frac{\mathrm{d}E}{\mathrm{d}B}\frac{\mathrm{d}B}{\mathrm{d}r}\Big|_{r=R_0} = -V\frac{\chi_1 B_0^2}{\mu_0 a}e^{-2R_0/a}$ (attractive). At detachment the normal force vanishes: $F = mg\cos\alpha + m\omega^2 D/2$.



So $\cos\alpha = \frac{\chi_1 B_0^2}{\mu_0 a \rho_1 g}e^{-2R_0/a} - \frac{\omega^2 D}{2g} > 1$ (eq. 2): the particle leaves at the lowest point, $\alpha = 0$. Neglecting D against d, the flight time is $t = \sqrt{2d/g}$ (since $(\omega D/g)^2 \ll d/g$), and

$$x = -\frac{\omega D}{2}\sqrt{\frac{2d}{g}} = -22.7\ \text{cm}.$$

Magnetic force $F_m = -\partial E/\partial r$	**0.1**
Result of B2 substituted: $F_m = -\frac{\chi_1 V B_0^2}{a\mu_0}e^{-2R_0/a}$	**0.2**
Detachment condition $\rho_1 g V\cos\varphi + F_m = \rho_1 V\omega^2(D/2 + R_0)$	**0.2**
Detachment at the lowest point	**0.2**
Formula $x = -\frac{\omega D}{2}\sqrt{2d/g}$	**0.2**
Value $x \approx -22$ cm	**0.2**

B.4
1.1pt

For different sizes, $\cos\alpha$ and the effective diameter change. For this range all particles still leave at the lowest point, and the effective diameter ranges over $[D + 1.8R_0,\ D + 2.2R_0]$. Adding the size of the particles themselves,

$$\Delta_0 = \left|\frac{\mathrm{d}x}{\mathrm{d}D}\right| \cdot 0.4R_0 + 2R_0 = 2.1\ \text{mm}.$$

Change of the magnetic force $\Delta(F_m/V) = -2\frac{F_m}{V}\frac{R_0}{a}\varepsilon$	**0.3**
Change of the centrifugal force $\Delta(F_c/V) = \rho_1\omega^2 R_0\varepsilon$	**0.2**
Detachment still at the lowest point	**0.2**
$\Delta x/x = 2\varepsilon R_0/D$	**0.2**
Formula $\Delta_0 = 2R_0 + 4x\varepsilon R_0/D$	**0.1**
Value $\Delta_0 = 2.1$ mm	**0.1**

B.5
1.1pt

From eq. (2) the smallest detachment angle obeys $\cos\alpha_{\min} = \frac{\chi B_0^2}{\mu_0 a \rho g} e^{-2R_0/a}$: 2.44 for kind 1 and 0.29 for kind 2, so kind 1 may leave anywhere while kind 2 leaves only for $\alpha \geq \arccos 0.29$. Take the separation $\Delta x = \Delta$ as the criterion. *Lower bound*: kind 1 leaves at the bottom, kind 2 at some α_2; the resulting condition has no solution below $\alpha_2 = \pi/2$, so $\frac{D}{2} + \frac{\omega D}{2}\sqrt{\frac{2d}{g}} = \Delta$ and

$$\omega_{\min} = \left(\frac{2\Delta}{D} - 1\right)\sqrt{\frac{g}{2d}} = 2\pi \cdot 190\ \mathrm{min}^{-1}.$$

Upper bound: kind 1 leaves at some α_1, kind 2 slightly lower. The condition has no solution in the approximation $(\omega D/g)^2 \ll d/g$, so the exact flight time (eq. 2 of B3) must be used; solving numerically gives $\alpha_1 = 0.629$ and

$$\omega_{\max} = 2\pi \cdot 270\ \mathrm{min}^{-1}.$$

Lower bound considered correctly (kind 2 leaving at an unknown point of the roller)	**0.3**
$\omega_{\min} = (2\Delta/D - 1)\sqrt{g/2d} = 2\pi \cdot 190\ \mathrm{min}^{-1}$	**0.2**
Upper bound considered correctly (kind 1 leaving at an unknown point)	**0.3**
Flight time of kind 1 taken into account correctly	**0.1**
$\omega_{\max} = 2\pi \cdot 270\ \mathrm{min}^{-1}$	**0.2**